

Regression (Chapter-8 2nd part)

The statistical tool with the help of which we are in a position to estimate (or predict) the unknown values of dependent variable from known values of independent variable is called **regression**. With the help of regression analysis, we are in a position to find out the average probable change in one variable give a certain amount of change in another. For example, if we know that two variables price (X) and demand (Y) are closely related we can find out the most probable value of X for a given value of Y or the most probable value of Y for a given value of X.

Regression equation of Y on X:

The regression equation of Y on X is expressed as follows:

$Y = a_1 + b_1 X$, where Y is the dependent variable to be estimated and X is the independent variable. In this equation a and b are two unknown constants (fixed numerical values) which determine the position of the line completely. The constants are called the parameters of the line. The parameter 'a₁' determines the level of the fitted line (i.e. the distance of the line directly above or below the origin). The parameter 'b₁' determines the slope of the line i.e. the change in Y for unit change in X. If the values of the constants 'a₁' and 'b₁' are obtained, the line is completely determined. 'a₁' and 'b₁' are obtained by least square method.

Here,

$$a_1 = \bar{y} - b_1 \bar{x} \text{ and } b_1 = \frac{\sum xy - \sum x \sum y / n}{\sum x^2 - (\sum x)^2 / n}$$

Regression equation of X on Y:

The regression equation of X on Y is expressed as follows:

$$X = a_2 + b_2 Y$$

'a₁' and 'b₁' are obtained by least square method.

Here,

$$a_2 = \bar{x} - b_2 \bar{y} \text{ and } b_2 = \frac{\sum xy - \sum x \sum y / n}{\sum y^2 - (\sum y)^2 / n}$$

Example:

A researcher want to find out if there is any relationship between the ages of the households and the ages of wives .He took a random sample of 7 couples whose respective ages are given below.

Age of husbands	25	27	29	32	35	37	39
Age of wives	18	20	20	25	25	30	37

- Compute the regression equation of age of wives on age of husbands.
- Compute the regression equation of age of husbands on age of wives.
- Determine the most likely age of husband for age of wife 35 years.
- Determine the most likely age of wife for age of husbands 36 years.

Solution:

Let, y Age of wives and x denotes age of husbands.

- Regression equation of age of wives (y) on age of husbands (x) is:

$$Y = a_1 + b_1 X$$

$$a_1 = \bar{y} - b_1 \bar{x} \quad \text{and} \quad b_1 = \frac{\sum xy - \sum x \sum y / n}{\sum x^2 - (\sum x)^2 / n}$$

Calculation of Regression coefficients

x	y	xy	x ²	y ²
25	18	450	625	324
27	20	540	729	400
29	20	580	841	400
32	25	800	1,024	625
35	25	875	1,225	625
37	30	1,110	1,369	900
39	37	1,443	1,521	1,369
$\sum x = 224$	$\sum y = 175$	$\sum xy = 5,798$	$\sum x^2 = 7,334$	$\sum y^2 = 4,643$

The regression equation of age of wives (y) on age of husbands (x) is expressed as follows:

$$Y = a_1 + b_1 X$$

$$\text{where, } b_1 = \frac{\sum xy - \sum x \sum y / n}{\sum x^2 - (\sum x)^2 / n} = \frac{5798 - 224 \times 175 / 7}{7334 - (224)^2 / 7} = \frac{5798 - 5600}{7334 - 7168} = \frac{198}{166} = 1.20$$

$$a_1 = \bar{y} - b_1 \bar{x} = \frac{\sum y}{n} - b_1 \frac{\sum x}{n} = \frac{175}{7} - 1.20 \times \frac{224}{7} = 25 - 1.20 \times 32 = 25 - 38.4 = -13.4$$

The regression equation of age of wives (y) on age of husbands (x) is expressed as follows:

$$Y = a_1 + b_1 X = -13.4 + 1.20x$$

ii) Regression equation of age of husbands (x) on age of wives (y) is:

$$\begin{aligned} X &= a_2 + b_2 Y \\ a_2 &= \bar{x} - b_2 \bar{y} \\ b_2 &= \frac{\sum xy - \sum x \sum y / n}{\sum y^2 - (\sum y)^2 / n} \end{aligned}$$

We have,
$$b_2 = \frac{\sum xy - \sum x \sum y / n}{\sum y^2 - (\sum y)^2 / n}$$

$$\frac{5798 - 224 \cdot 175 / 7}{4643 - (175)^2 / 7} = \frac{5798 - 5600}{4643 - 4375} = \frac{198}{268} = 0.74$$

$$a_2 = \bar{x} - b_2 \bar{y} = \frac{\sum x}{n} - b_2 \frac{\sum y}{n} = \frac{224}{7} - 0.74 \times \frac{175}{7} = 32 - 0.74 \times 25 = 32 - 18.5 = 13.5$$

Therefore, Regression equation of age of husbands (x) on age of wives (y) is:

$$X = a_2 + b_2 Y = 13.5 + 0.74y$$

iii) the most likely age of husband for age of wife 35 years.

$$\begin{aligned} X &= a_2 + b_2 Y \\ &= 13.5 + 0.74 \cdot 35 \\ &= 13.5 + 25.9 = 39.4 \end{aligned}$$

iv) The most likely age of wife for age of husbands 36 years.

$$\begin{aligned} Y &= a_1 + b_1 X \\ &= -13.4 + 1.20 \cdot 36 \\ &= -13.4 + 43.4 = 29.8 \end{aligned}$$

Exercise

The following table shows the the age and blood pressure of 8 persons

Age	50	63	45	36	72	65	47	25
Blood pressure	62	60	51	25	79	43	60	33

- Compute the regression equation of blood pressure on age .
- Compute the regression equation of age on blood pressure .
- Find out the expected blood pressure of a person who is 52 years .
- Find out the expected age of a persons whose blood pressure is 70.